ML4VA PROJECT: THRESHOLD-BASED POLICY MAKING FRAMEWORK FOR VIRGINIA HEALTH OPPORTUNITY

UVA CS 4774 MACHINE LEARNING COURSE PROJECT

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ABSTRACT

Our project employs machine learning to advance health opportunities in Virginia by predicting and improving the Health Opportunity Index (HOI) across various communities. We developed a regression-focused neural network model, optimized through hyperparameter tuning, to capture the nuanced interplay between health determinants. Given the lack of extensive validation datasets, we innovatively leveraged partial dependence plots based on the Random Forest Regression Model for insightful threshold-based policy development, prioritizing actions where they are most needed. This framework serves as a preliminary step towards a data-driven public health strategy, with future work set to incorporate broader data for model refinement and to ensure adaptability to real-world applicability.

1 Motivation

The motivation behind our project is to harness the predictive power of machine learning to tackle public health disparities across Virginia. Recognizing that health opportunities vary widely by location, our goal is to analyze and interpret complex health data to inform targeted, effective policy interventions. By developing a machine learning framework that can accurately identify key factors influencing health outcomes, we aim to provide actionable insights to policymakers, ultimately contributing to the improvement of health equity and quality of life for all Virginians.

2 Dataset

Dataset: Virginia Health Opportunity Index

Description: The Virginia Health Opportunity Index (HOI) is a group of indicators that provide broad insight into the overall opportunity Virginians have to live long and healthy lives based on the Social Determinants of Health. It is a hierarchical index that allows users to examine social determinants of health at multiple levels of detail in Virginia. It is made up of over 30 variables, combined into 13 indicators, grouped into four profiles, which are aggregated into a single Health Opportunity Index. The dataset encompasses various indicators reflecting the overall opportunity for Virginians to live long and healthy lives based on Social Determinants of Health.

3 Related work

Shelley White-Means' paper "Intervention and Public Policy Pathways to Achieve Health Care Equity," reviews diverse strategies to achieve health care equity, guided by the National Institute of Minority Health and Health Disparities (NIMHD) framework. It underscores the need for interventions and policy changes at multiple levels, including individual, community, and societal, to address the multifaceted nature of health disparities. The critiques point to a lack of additional data to further test the approaches and call for future research to fill gaps in understanding the interplay of socio-environmental factors and health outcomes. The paper advocates for a comprehensive, multifaceted approach to healthcare equity, recognizing the complexity of healthcare disparities and the necessity for persistent, dynamic efforts in public health policy and research. This work serves as a methodological guide for our project in employing rigorous methodologies to study the health opportunities in Virginia.

4 Method

4.1 Data preprocessing and feature selection

Our methodology encompassed several steps, starting with data preprocessing and feature selection. This initial step is crucial to ensure the subsequent algorithms operate at peak efficiency. Following this, we employed clustering algorithms, specifically the K-means algorithm, to segment environmental data. The choice of K-means was based on its capability to discern patterns or groups within datasets, an aspect that greatly aids in subsequent analytical processes.

4.2 Health Opportunity Index prediction

For predictive modeling, we explored multiple regression methods: Linear Regression, Random Forest, and Decision Tree. By employing a range of regression techniques, we were able to comparatively evaluate their performance, an approach that lends robustness to our analytical methodology.

4.3 Policy suggestion

For our objective of formulating policy recommendations for communities, we currently lack specific data in this domain. Consequently, our approach involves employing a threshold-based method to facilitate policy recommendations.

To achieve this, we employ various techniques, including Artificial Neural Network (ANN), Random Forest, and Shap analysis. These methods help us assess the "importance" of different features in relation to health accessibility. For each feature, we endeavor to establish a threshold that maximally influences health accessibility and then recommend communities to surpass this threshold. This strategy ensures that communities fully leverage the potential of each feature and allocate limited resources to those most impactful features.

5 Preliminary Experiments

5.1 Data Pre-Processing and Feature Selection

Our analysis commenced with data cleansing, where we addressed any missing or anomalous entries to ensure the integrity of our subsequent analyses. This foundational step was pivotal to ensure the reliability of the models and methods applied.

5.2 Environmental Segmentation

We employed the K-means clustering algorithm to achieve this. The clusters identified by K-means can reveal patterns or subgroups within the data. The elbow method guided us in determining the optimal number of clusters. The elbow method indicated the ideal K value (number of clusters) to be 3. Based on this, we segmented our dataset into three distinct clusters, each rooted in the environmental indicator - air quality. A dimensionality reduction was executed using Principal Component Analysis (PCA) before clustering with the K-means algorithm.

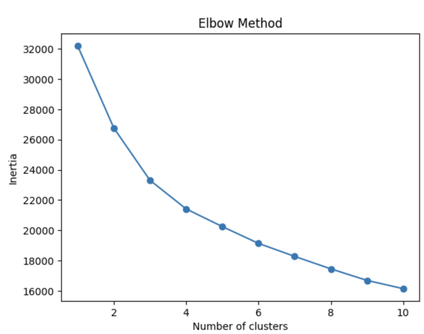


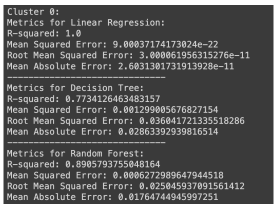
Figure 1: Elbow Method

Figure 2: 2D PCA Plot

PCA's goal was to maximize variance capture within the initial components. The clusters were labeled as Cluster 0 (red), Cluster 1 (blue), and Cluster 2 (green). The evident distinction between clusters indicates successful data groupings by the K-means algorithm based on specific health metrics or features.

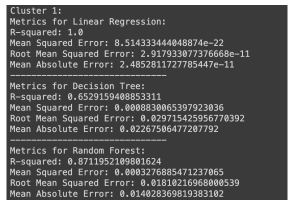
5.3 Regression Analysis

To predict healthcare accessibility, we experimented with multiple regression methods with environmental factors serving as our primary explanatory variables. While several regression methods were tested, Random Forest emerged as the method of choice due to its superior performance and ability to handle complex data structures. We utilized multiple metrics such as R-squared value, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) to gauge the model's accuracy and reliability. A noteworthy observation was the unusually high R-squared value of 1.0 for Linear Regression across all clusters, typically indicative of overfitting. To further investigate, we compared the model's training set performance against the test set. The similar performance metrics dispelled our concerns about overfitting. However, Decision Trees and Random Forests displayed lower R-squared values than Linear Regression, suggesting a more conservative variance capture. This could indicate a more realistic model fit, not swayed by potential overfitting.



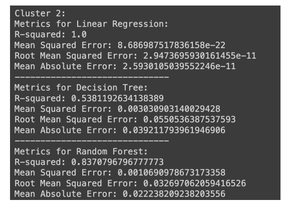
```
Cluster 0:
Metrics for Linear Regression:
R-squared: 1.0
Mean Squared Error: 9.00037174173024e-22
Root Mean Squared Error: 3.000061956315276e-11
Mean Absolute Error: 2.6031301731913928e-11
------------------------------
Metrics for Decision Tree:
R-squared: 0.7734126463483157
Mean Squared Error: 0.001299005676827154
Root Mean Squared Error: 0.036041721335518286
Mean Absolute Error: 0.02863392939816514
------------------------------
Metrics for Random Forest:
R-squared: 0.8905793755048164
Mean Squared Error: 0.0006272989647944518
Root Mean Squared Error: 0.025045937091561412
Mean Absolute Error: 0.01764744945997251
```

Figure 3: Cluster 0 Results



```
Cluster 1:
Metrics for Linear Regression:
R-squared: 1.0
Mean Squared Error: 8.514333444048874e-22
Root Mean Squared Error: 2.917933077376668e-11
Mean Absolute Error: 2.4852811727785447e-11
------------------------------
Metrics for Decision Tree:
R-squared: 0.6529159408853311
Mean Squared Error: 0.0008830065397923036
Root Mean Squared Error: 0.029715425956770392
Mean Absolute Error: 0.02267506477207792
------------------------------
Metrics for Random Forest:
R-squared: 0.8711952109801624
Mean Squared Error: 0.0003276885471237065
Root Mean Squared Error: 0.0181021696800539
Mean Absolute Error: 0.014028369819383102
```

Figure 4: Cluster 1 Results



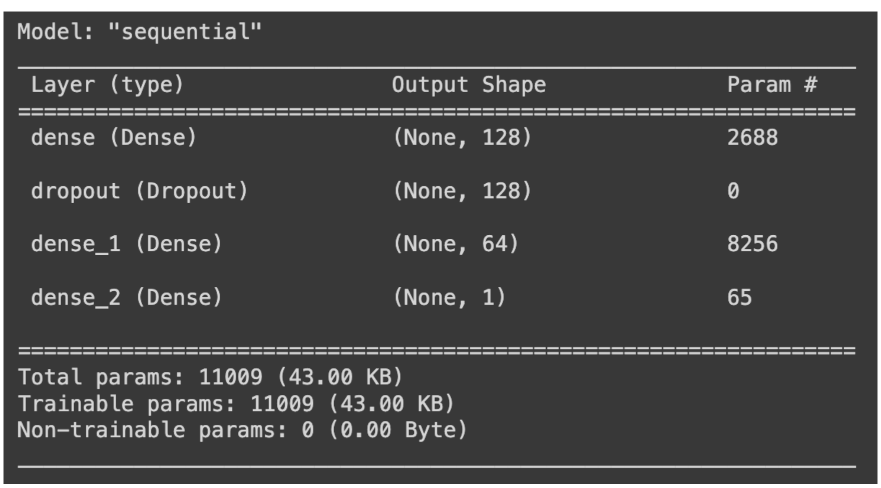
```
Cluster 2:
Metrics for Linear Regression:
R-squared: 1.0
Mean Squared Error: 8.686987517836158e-22
Root Mean Squared Error: 2.9473695930161455e-11
Mean Absolute Error: 2.5930105039552246e-11
------------------------------
Metrics for Decision Tree:
R-squared: 0.5381192634138389
Mean Squared Error: 0.003030903140029428
Root Mean Squared Error: 0.0550536387537593
Mean Absolute Error: 0.039211793961946906
------------------------------
Metrics for Random Forest:
R-squared: 0.8370796796777773
Mean Squared Error: 0.0010690978673173358
Root Mean Squared Error: 0.032697062059416526
Mean Absolute Error: 0.022238209238203556
```

Figure 5: Cluster 2 Results

Given these outcomes, while Linear Regression might harbor unresolved issues, Random Forest consistently outshone Decision Tree, showcasing its ensemble methodology's potency in minimizing prediction variance. The heightened prediction errors for Cluster 2 suggest potential noise or diminished predictive quality of features within this cluster. Conclusively, our choice for the primary regression technique for this project is the Random Forest algorithm.

5.4 Regression Neural Network Model Development

In addition to the application of the Random Forest algorithm, our approach includes the implementation of a specialized neural network, which is explicitly tailored for regression tasks. The rationale for training a neural network from scratch instead of implementing a pre-trained model is based on the fact that the majority of pre-trained models are inherently structured for classification purposes. In the context of regression tasks, it is typically more efficacious to train a neural network from the ground up, as this allows for a more targeted and relevant model development. During the training process, we experimented with different batch sizes and numbers of epochs. Here is the finalized neural network. After training the model, the results indicate that our neural network model is learning and improving its predictions



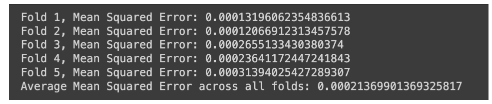
```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense (Dense)               (None, 128)               2688

 dropout (Dropout)           (None, 128)               0

 dense_1 (Dense)             (None, 64)                8256

 dense_2 (Dense)             (None, 1)                 65

=================================================================
Total params: 11009 (43.00 KB)
Trainable params: 11009 (43.00 KB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

Figure 6: Neural Network

over time, with relatively low error rates. The model does not show significant signs of overfitting, and the RMSE of 0.013400373129133383 suggests that the model fits the data well. Cross-validation is also essential to assess the generalizability of our model. We implemented the K-Fold Cross-Validation to our neural network model. The result of the K-Fold Cross-Validation has shown the robustness of our model.

```
Epoch 99/100
38/38 [==============================] - 0s 5ms/step - loss: 2.7977e-04 - root_mean_squared_error: 0.0167 - val_loss: 1.8696e-04 - val_root_mean_squared_error: 0.0137
Epoch 100/100
38/38 [==============================] - 0s 5ms/step - loss: 3.8766e-04 - root_mean_squared_error: 0.0197 - val_loss: 1.9657e-04 - val_root_mean_squared_error: 0.0140

[ ] test_loss = model.evaluate(X_test, y_test)
    print('Test MSE:', test_loss)

12/12 [==============================] - 0s 3ms/step - loss: 2.0059e-04 - root_mean_squared_error: 0.0142
Test MSE: [0.00020059205417055637, 0.014163052663207054]

[ ] rmse = np.sqrt(1.7957e-04)
    print("RMSE:", rmse)

RMSE: 0.013400373129133383
```

Figure 7: Neural Network Training Result



```
Fold 1, Mean Squared Error: 0.00013196062354836613
Fold 2, Mean Squared Error: 0.00012066912313457578
Fold 3, Mean Squared Error: 0.0002655133430380374
Fold 4, Mean Squared Error: 0.00023641172447241843
Fold 5, Mean Squared Error: 0.00031394025427289307
Average Mean Squared Error across all folds: 0.00021369901369325817
```

Figure 8: K-Fold Cross-Validation

5.5 SHAP Analysis

The main reason for constructing and training such a neural network model is to implement an analytical method known as SHAP (SHapley Additive exPlanations) Analysis. This technique, rooted in game theory, is employed to elucidate the outputs of machine learning models. The accompanying plot illustrates the features ranked in descending order of their importance to the model's output. This ranking is significant, as it highlights the feature that most substantially influences the model's predictions, positioned at the top of the plot. This analysis will benefit our project in the context of policy decision-making. Often, we encounter scenarios where multiple features simultaneously break predetermined thresholds, necessitating a decision on which feature requires prioritization for improvement. SHAP Analysis provides a robust framework for making such determinations, allowing for a more informed and effective approach to enhancing the pertinent features.

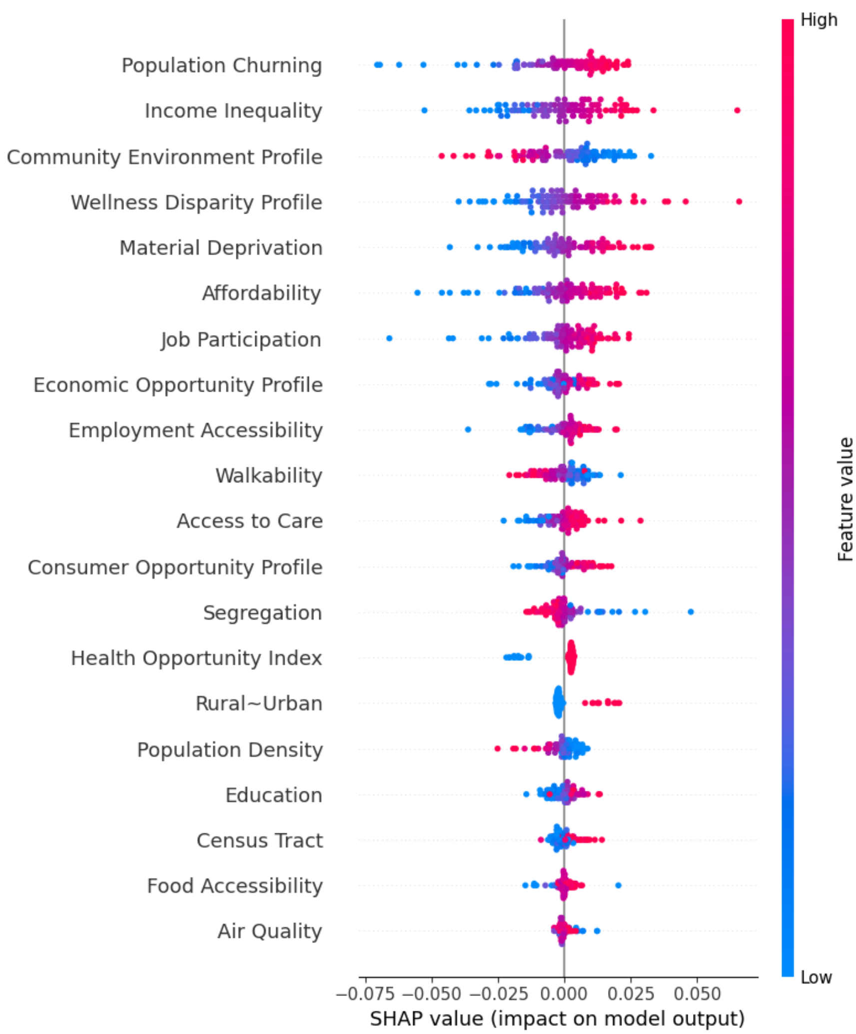


Figure 9: SHAP Analysis Summary

5.6 Random Forest Partial Dependence Result

Referring back to the Random Forest model previously discussed, we can use its determination of the partial dependence of the 'HOI' variable on various features to construct our Threshold-Based Policy-Making Framework. The graphical representation of the three features' dependencies as examples are shown below. The first plot below presents a pronounced steep slope near its center. This characteristic implies that focused enhancements in this specific area are likely to be highly beneficial. As a result, we set a threshold at this juncture for the 'Material Deprivation' feature. Conversely, some graphs, as illustrated in the second plot below, may feature a downward-sloping curve. This pattern suggests that, for certain variables, the values within the communities should be maintained below a predetermined threshold. Moreover, in situations where a clear threshold is not apparent, as shown in the third plot, which features a smoothly increasing curve (* It can also be a smoothly decreasing curve), we choose the median value of that feature as a provisional threshold. This median value will be identified as a 'secondary feature,' suggesting that the community's focus on this aspect should be secondary, contingent upon fulfilling criteria for more pivotal features identified as 'significant'.

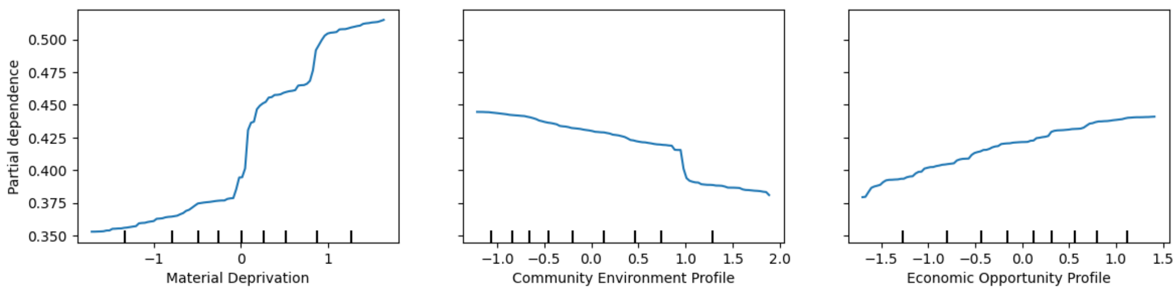


Figure 10: Partial Dependence of three example features

6 Threshold-Based Policy Making Framework Recommendations

6.1 Threshold-Based Policy Making Framework Result

By finalizing the result of Partial Dependence based on Random Forest, we construct the Threshold-Based Policy Making Framework. The framework uses the thresholds shown below to identify priority areas for improvement in different communities. For example, if a community exceeds the threshold for Material Deprivation, initiatives could be directed to address overcrowding or unemployment first. In situations where multiple thresholds are breached simultaneously, the framework can prioritize interventions based on the identity signs like 'significant' or 'secondary' and SHAP analysis which illustrates the importance of each feature on the Health Opportunity Index. Significant features are addressed first, followed by secondary ones. If several significant or secondary features require attention, SHAP analysis will guide policymakers to focus on the most impactful feature of the Health Opportunity Index.

Access to Care:	>0.01	secondary
Employment Accessibility:	>0.30	secondary
Affordability:	>0.36	significant
Air Quality:	>0.57	secondary
Population Churning:	>-0.51	significant
Education:	>0.48	secondary
Food Accessibility:	>-0.26	secondary
Income Inequality:	>0.17	significant
Job Participation:	>0.52	secondary
Population Density:	<-0.51	secondary
Segregation:	<0.10	secondary
Material Deprivation:	>0.02	significant
Walkability:	<0.51	secondary
Community Environment Profile:	<0.18	significant
Consumer Opportunity Profile:	>0.29	secondary
Economic Opportunity Profile:	>0.05	significant
Wellness Disparity Profile:	>-0.29	secondary

Figure 11: Final Threshold Result

By setting clear priorities based on the Random Forest model's insights, resources can be allocated more effectively to improve community health outcomes. Policymakers can create targeted interventions to address the most impactful areas, leading to more efficient use of funds and efforts. Communities can track progress with quantifiable metrics, ensuring transparency and accountability in health opportunity initiatives.

6.2 Conclusion: Limitation and Future work

This project's primary limitation lies in the lack of additional data for validation, restricting our ability to test the model's robustness further. Future work will focus on acquiring more diverse datasets to reinforce our findings. We also aim to explore the integration of real-time data analytics, enhancing our model's predictive accuracy and enabling dynamic policy adjustments. These steps will ensure continuous improvement in our framework's applicability to Virginia's evolving health landscape.

7 Project Code Link

Here is the link to our Google Colab Code:
`https://colab.research.google.com/drive/1ir3qrOcbfoDkwUA41O5CVJkAW3Y_kKJk?usp=sharing`

8 Member Contribution

- Zhiyuan Song: Discuss the possibility of cluster groups based on environmental indicators. Handle part of the data pre-processing and feature selection. Contributed to K-means clustering and gave initial establishment of regression analysis. Create and train the Neural Network. Conduct SHAP Analysis Summary. Introducing Random Forest to help generate Partial Dependence for threshold. Write most of the Video Presentation script and edit the video as post-production. Edited a large amount of the overleaf.
- Zihan Mei: Handled part of the data pre-processing. Implemented linear regression, decision tree, and random forest Algorithms for each cluster generated by K-means. Conducted Random Forest algorithm to analyze Partial Dependence. Generated Policy recommendation using threshold-based analysis. Edited a medium amound of the overleaf.
- Liran Li: Discuss the possibility of cluster groups based on environmental indicators. Handle part of the data pre-processing, conducted feasibility experiments with k-means and linear regression, helped analyze the RMSE of the data, made part of the decision tree. Responsible for integration and debugging of the code in the submitted version, responsible for editing the overleaf in the submitted version.

9 Reference

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